Customer Segmentation using K-Means (Python Code)

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# Customer Segmentation using K-Means
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

data = {
'Annual_Income':[15,16,17,18,25,30,35,40,45,50,55,60,65,70,75,80,85,90],
'Spending_Score':[39,81,6,77,40,76,6,94,3,72,14,60,20,80,5,90,15,95]}

df = pd.DataFrame(data)
kmeans = KMeans(n_clusters=3, random_state=42)
df['Cluster']=kmeans.fit_predict(df[['Annual_Income','Spending_Score']])

plt.figure(figsize=(8,6))
plt.scatter(df['Annual_Income'],df['Spending_Score'],c=df['Cluster'],cmap='viridis',
s=100)
centroids=kmeans.cluster_centers_
plt.scatter(centroids[:,0],centroids[:,1],color='red',marker='X',s=250,label='Centro
ids')
plt.title('Customer Segmentation using K-Means')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score')
plt.legend()
plt.grid(True)
plt.show()
```